

No Heroes No Lies

Backend Game Rules Reference — Document 1

v1.0 — For AI-assisted backend development

1. Game Overview

No Heroes No Lies is a realtime multiplayer card game of deception and combat. Players conceal their true hero identity, bluff opponents, fight monsters, and use powers to accumulate coins. The game is built around a public declaration system where declared identities trigger passive effects even if false, creating a meta-game of trust, accusation, and timing.

2. Player Resources

Each player tracks three independent resources throughout the game:

Resource	Starting Value	Purpose	Loss Condition
Life	3	Survival — game over at 0	Failed monster fight, false accusation penalty, targeted powers
Coins	0	Win condition — reach players+1 to win	Removed by powers (remove_coin, steal_coin)
Gems	0	Power fuel — spent to activate paid powers	Spent on powers, stolen by powers, lost via all_in

3. Game Setup

3.1 Lobby & Start Condition

- A lobby is created by the Game Master (the player who created the session).
- The game starts when the Game Master manually triggers start AND all player slots are filled.
- The server must reject start attempts if any slot is unfilled.

3.2 Initialization Sequence

The server executes the following steps in strict order upon game start:

1. Shuffle the full hero deck.
2. Draw 1 hero card at random and add it to the burned pile (permanently removed for this session).
3. Deal 1 hero card to each player (private — never broadcast to other players).
4. Shuffle the monster deck according to rarity spawn rates.
5. Draw 2 monsters from the monster deck and place them in the active monster slots.
6. Determine the first player: randomly if first session, otherwise last session's winner goes first.
7. All player identities begin as UNKNOWN.

NOTE

The burned card is known to no player. Its ID is server-only and must be excluded from shuffle_heroes_deck power resolution.

4. Turn Structure

Turns proceed in a fixed player order. Each turn follows a strict phase sequence. The server enforces valid actions per phase and rejects anything outside the current phase.

Phase	Description	Who Acts
TURN_START	Resolve passive triggers that fire at turn start (e.g. Werewolf alternate_strength)	Server (automatic)
IDENTITY_DECLARATION	Active player declares their identity (true or false hero ID)	Active player
ACCUSATION_WINDOW	Other players may accuse the active player of lying. Window stays open until action is committed or an accusation resolves.	Any other player
ACTION_PHASE	Active player selects and executes one of three action types	Active player
PASSIVE_RESOLUTION	Server resolves all passive effects triggered by the action	Server (automatic)
TURN_END	Check win conditions. Advance to next player.	Server (automatic)

5. Identity Declaration

5.1 Rules

- The active player must declare an identity at the start of their turn before acting (except for the Demask action which requires no declaration).
- The declared identity can be any hero ID — it does not need to match the player's true hero card.
- A declaration is stored as `declaredIdentityId` on the player state and broadcast to all players.
- The declaration is sticky: it persists until the player declares again on a future turn.
- A player whose bluff was caught has their identity reset to UNKNOWN and must redeclare next turn.

5.2 Identity State Values

State	Meaning
UNKNOWN	Player has not yet declared this session, or their last bluff was caught
<herold>	Player has declared a specific hero identity (may or may not be true)

IMPORTANT

Passive powers run based on declared identity, not true identity. If a player declared Chevalier (even falsely), the teamwork passive fires when anyone gains gems — until the bluff is caught or they redeclare.

6. Accusation / Challenge Window

6.1 Trigger & Timing

- The accusation window opens immediately when the active player declares their identity.
- The window remains open until either: (a) the active player commits their action, or (b) an accusation is made and resolved — whichever comes first.
- Once one accusation resolves, the window closes immediately. All other pending accusations are discarded.
- First accusation received by the server wins. Simultaneous accusations are resolved by arrival order.

6.2 Accusation Outcomes

Scenario	Result
Accusation correct — player was bluffing	Accused loses 1 life. Accused draws new hero immediately. Accused identity resets to UNKNOWN. Active player's action is cancelled. Accused's turn ends.

Scenario	Result
Accusation wrong — player was truthful	Accuser loses 1 life. Action proceeds normally.
No accusation before action committed	Window closes. Action proceeds.

6.3 Accusation Exemptions

The following situations do NOT trigger a false-accusation life penalty for the accuser:

- Accusing during an Exorcist shoot_player power use — the power's own resolution handles consequences.
- The Demask action — this action bypasses identity declaration entirely and has no accusation window.

6.4 Exorcist Accusation Edge Case

Special stacking scenario: If player A declares being an Exorcist and uses shoot_player against player B, and player B then accuses player A of lying while player A is genuinely an Exorcist:

- The shoot_player power resolves normally (A was truthful).
- The false accusation penalty applies (B accused wrongly).
- Player B loses 2 lives total: 1 from shoot_player resolution + 1 from false accusation.

7. Player Actions

On their turn, the active player must choose exactly one of three action types. The Demask action is only available if the player has 6 or more gems.

7.1 Action A — Pick Card + Power

The full sequence for this action is:

8. Declare identity (triggers accusation window).
9. Draw the top card from the hero deck.
10. Discard either the drawn card or the player's current hero card. The discarded card is publicly visible.
11. Select a power to use (free or paid). If no power is desired, the action ends here.
12. If a paid power is selected: the hero's free power (Power 1) triggers first and must be resolved before the paid power (Power 2) executes.

**POWER
SEQUENCE**

Selecting a paid power is a two-step commit: Power 1 fires automatically and player must complete it, then Power 2 executes. The server must gate Power 2 execution behind Power 1 completion.

7.2 Action B — Attack Monster

The full sequence for this action is:

13. Declare identity (triggers accusation window).
14. Select one of the two active monsters to attack.
15. Server compares declared identity's strength against the monster's strength.
16. If declared strength > monster strength: active player gains monster's loot (coins + gems). Passive loot effects run (e.g. Chevalier teamwork).
17. If declared strength <= monster strength: active player loses 1 life.
18. Slain monster is removed from active slots. A replacement monster spawns at the start of the next turn (not immediately).

**STRENGTH
NOTE**

Strength is derived from the declared identity, not the true hero card. A player declaring Chevalier fights as strength 11 regardless of their real card.

7.3 Action C — Demask (requires 6+ gems)

This action does NOT require identity declaration. The accusation window does not open.

19. Player selects a target (any other player).
20. Player guesses the target's true hero card ID.
21. If guess is correct: target loses their hero card (draws a new one immediately) and loses 1 life.
22. If guess is wrong: nothing happens to the target. (No life penalty for the demask attacker.)
23. The 6 gems are NOT spent on demask — they are only the unlock threshold.

8. Passive Power Resolution

After any action that generates gems, coins, or triggers a specific event, the server runs a passive resolution pass over all players' declared identities.

Passive	Hero	Trigger	Effect	Stacking Order
teamwork	Chevalier	gems_gained (any player)	The player who gained gems doubles the amount	1st — applied before tribute
tribute	Mad Prince	gems_gained (other player)	Half of another player's gem gain is redirected to Mad Prince	2nd — applied after teamwork doubling
keep_gems	Geant	steal_attempt targeting this player	Cancels the steal entirely	Immediate — blocks before steal resolves
alternate_strength	Werewolf	turn_start	Strength alternates between 5 and 12 each turn	Automatic — server updates on TURN_START phase
royal_immunity	Mad Queen	always	Cannot be targeted by any attack power	Pre-check — server rejects targeting before power resolves

PASSIVE STACKING EXAMPLE

Chevalier declared + Mad Prince declared, another player gains 4 gems from monster loot: teamwork fires first (4 becomes 8), then tribute fires (Mad Prince takes 4, loot player keeps 4).

9. Discard Pile & Visibility

- The last discarded hero card is always publicly visible to all players.
- Only one card is in the 'visible discard' slot at a time — it is replaced when a new card is discarded.
- Previous discards are in a non-visible discard pile (still accessible for powers like `fight_player_with_discarded_card` and `command_the_dead`).
- Burned cards are tracked separately and are never returned to any deck or pile.

10. Monster Management

- There are always 2 active monster slots.
- At game start, 2 monsters are drawn and placed in the slots.
- When a monster is slain, its slot becomes empty for the remainder of that turn.
- At the start of the next turn, before any action, the server fills empty slots by drawing from the monster deck.
- If the monster deck is exhausted, reshuffle all non-active monsters back into a new deck.

11. Win Conditions

The server checks win conditions at the end of every turn, in this priority order:

Condition	Winner	Check Timing
A player's life reaches 0	Last surviving player wins	Immediately when life hits 0, not just at turn end
Only 1 player remains alive	That player wins	End of turn
A player's coin total equals (number of players + 1)	That player wins	End of turn, after all loot and passives resolve

COIN WIN NOTE

Win coin target = total players at game start + 1. It does not adjust if players are eliminated mid-game.

12. Player Elimination

- A player is eliminated when their life reaches 0.
- Eliminated players are removed from the turn order immediately.
- Their hero card is added to the discard pile (not burned).
- Their declared identity is cleared.
- Resources (gems, coins) are removed from the game — not redistributed.
- If only 1 player remains after elimination, the game ends immediately.

13. Special Card Rules

13.1 Ghost of the King (Mythic Monster)

- 0.5% spawn rate — treated as a special case in monster deck generation.

- Cannot be the target of a challenge/accusation window (no window opens when fighting it).
- Only Mad Prince (strength 13) and Werewolf in wolf form (strength 12) can realistically defeat it (strength 15).
- Mad Prince / Mad Queen vengeance bonus: +2 coins, +5 gems on defeat.
- Mad Queen command_the_dead bonus: +1 coin, +3 gems when using a discarded hero against it.

13.2 Mad Queen

- Royal Immunity passive is always active — server must reject any attack power targeting her (parry_this_you_f_casual, shoot_player, steal_gem, steal_coin, etc.).
- Gets no loot from direct personal monster fights.
- Must use command_the_dead to generate resources from monsters.

13.3 Werewolf

- True strength is tracked as a parity value (even turn = human form strength 5, odd turn = wolf form strength 12, or vice versa — define parity at session init).
- The server updates the Werewolf's effective strength at the TURN_START phase automatically.
- When a player is the Werewolf and declares identity as Werewolf, their strength reflects the current form.

13.4 Shapeshifter

- mimic_hero: copies the target's true hero abilities for the current turn.
- mimic_power: copies and executes the target's Power 1 (order-0 active power).
- If Shapeshifter mimics Mad Queen, royal_immunity does NOT transfer — only active powers copy, not permanent passives.

14. Server-Side Validation Rules

The server must enforce all of the following before executing any game action:

Rule	Check	Reject Reason
Phase gate	Action type must be valid for current phase	INVALID_PHASE
Active player gate	Only the active player can submit actions	NOT_YOUR_TURN

Rule	Check	Reject Reason
Gem cost	Player must have $\geq$ power gem cost before power resolves	INSUFFICIENT_GEMS
Demask threshold	Player must have $\geq$ 6 gems to attempt demask	INSUFFICIENT_GEMS_FOR_DEMASK
Royal immunity	Mad Queen cannot be targeted by attack powers	TARGET_IMMUNE
Strength comparison	Computed server-side from declared identity strength	CLIENT_CANNOT_RESOLVE
Burn pile exclusion	Burned heroes excluded from all decks and shuffle_heroes_deck	CARD_IS_BURNED
Ghost King no-challenge	Do not open accusation window for Ghost King fights	SPECIAL_MONSTER_RULE
Dead player actions	Eliminated players cannot submit actions	PLAYER_ELIMINATED
Self-targeting	Most targeted powers cannot target self (validate per power)	INVALID_TARGET

15. Open Questions for Future Specification

The following mechanics require explicit ruling before implementation:

- Does Shapeshifter's mimic_hero copy declared or true identity of the target?
- What happens when two passives conflict in resolution order beyond teamwork + tribute?
- Is there a turn timer? If yes, what happens on timeout (auto-pass, auto-random action)?
- Can a player choose not to use a power after picking a card (action A with no power)?
- What is the minimum player count? Does win coin target change at 2 players?